UNIT 2 · 9 hours

Losses and Efficiency of a Transformer



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Identify core and copper losses
- Explain hysteresis and eddy-current loss
- Calculate efficiency at any load and power factor
- State the condition for maximum efficiency

Syllabus context

Transformers: necessity of transformers, principle of operation, types and construction of single-phase transformers, EMF equation, losses and efficiency.

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Losses in a transformer

Core (iron) loss Pi

Hysteresis + eddy current. Depends on voltage and frequency; practically constant from no load to full load.

Copper loss Pc

$I1^2R1 + I2^2R2$ in the windings. Varies as the square of load current.

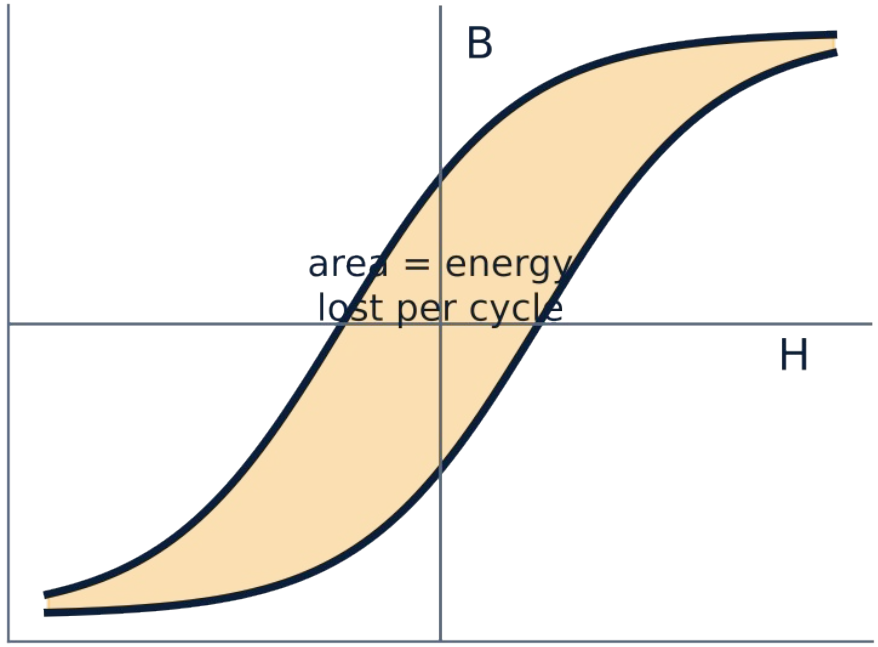
Others

Stray load loss and dielectric loss: small.

Total loss = Pi + Pc; no rotating parts, so no friction loss.

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Hysteresis loss



- B lags H in iron: the B–H curve forms a loop.
- Area of the loop = energy lost per cycle per unit volume.
- Loss ∝ f × area: silicon steel has a thin loop.
- Eddy-current loss ∝ f^2 Bm^2 t^2 (t = lamination thickness).

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Efficiency

$$\eta = \text{Output} / \text{Input}$$

= Output / (Output + losses)

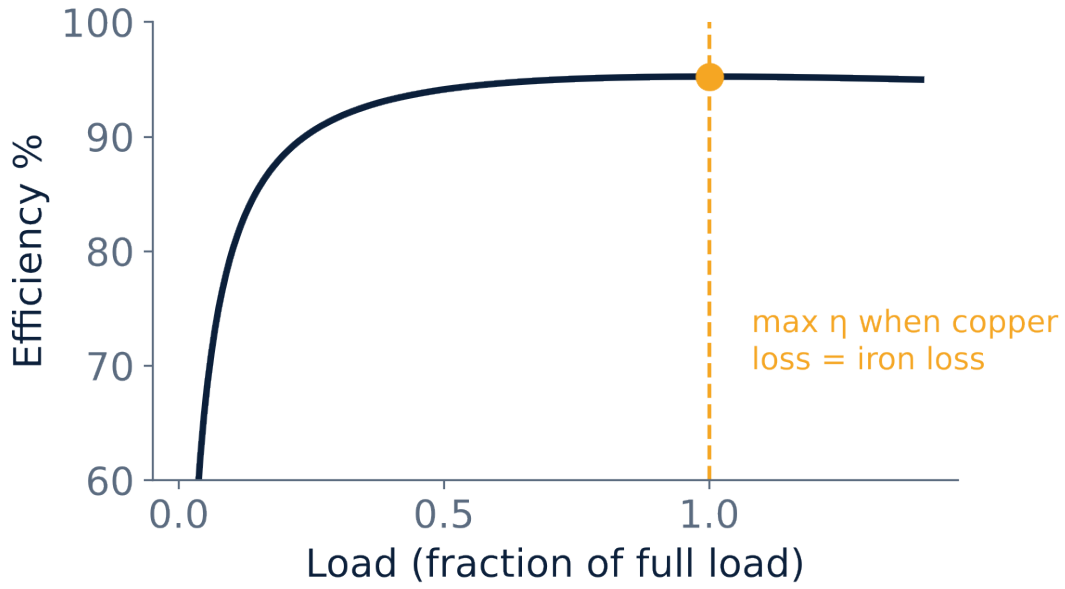
$$\eta = x\,S\cos\varphi \,/\, (x\,S\cos\varphi + Pi + x^2 Pc)$$

x = fraction of full load, S = kVA rating

- Typically 95–99 % for power transformers.
- Maximum efficiency when copper loss = iron loss: $x = \sqrt{(Pi/Pc)}$.
- Distribution transformers are designed for maximum efficiency at ≈ 60–70 % load.

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Efficiency versus load



- Efficiency rises quickly with load.
- Peak occurs where Pc = Pi.
- Falls slightly above that load as Pc grows with x^2.

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Numerical: full-load efficiency

Problem A 25 kVA, 2000/200 V transformer has iron loss 350 W and full-load copper loss 400 W. Find efficiency at full load, 0.8 pf lagging.

- Output = 25 × 0.8 = 20 kW
- Total loss = 0.35 + 0.4 = 0.75 kW
- $\eta = 20/(20 + 0.75) \times 100$

Answer: η = 96.38 %

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Numerical: maximum efficiency

Problem For the same transformer find the load at which efficiency is maximum and the maximum efficiency at unity power factor.

- $x = \sqrt{(Pi/Pc)} = \sqrt{(350/400)} = 0.935$
- Output = 0.935×25 kVA $\times 1 = 23.4$ kW
- Losses = 2×350 W = 0.7 kW
- $\eta = 23.4/(23.4 + 0.7)$

Answer: x = 0.935 (23.4 kVA), ηmax = 97.1 %

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Key takeaways

- Iron loss is constant; copper loss $\propto$ (load)^2
- Hysteresis loss depends on the B–H loop area; eddy loss on lamination thickness
- η = output / (output + Pi + Pc)
- Maximum η when Pc = Pi

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Practice questions

Q1. List the losses in a transformer and how each is reduced.

Q2. A 100 kVA transformer has Pi = 1 kW and Pc = 1.5 kW (full load). Find η at full load, unity pf.

Q3. Find the load fraction for maximum efficiency in question 2.

Q4. Why is the core loss assumed constant at all loads?